

Short Answer Test

1) Hospital - Data warehouse

a) show please schema

Patient - Dim		visit - Fact	Doctor - Dim
Patient - keys	PK	visit - key	Doctor keys
Patient - ID	FK	Patient - keys	Doctor - ID
Patient - Name	FK	Doctor - keys	Doctor - Name
Age			

b) OLAP operations

Starting cuboid

[Patient, Doctor, Department, time]

- * Roll NO Patient → All
- * Roll NO Doctor → All
- * Roll NO Time → year
- * Keep Department level unchanged
- * Aggregate using Average (Avg-charge)

Result:

[All, All, Department, year]

c) number of cuboids

Formula:

NO of cuboids = n (level of each dimension)

There are dimensions

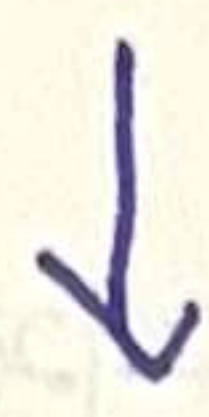
Each has 4 level

$$4 \times 4 \times 4 \times 4$$

= 256 cuboids

2) Smart city star schema

Time - Dim



Traffic - Dim ⇒ smart city - Fact

Transport - Dim

Location - Dim



Citizen - Dim

3) Banding Data warehouse

a) Grala as schema

Brand - Dim



Transaction - Fact →

account - Type - Dim

measures:

Transaction count
Amount

Customer - Dim



Branch - Dim → Loan - Fact —

Time - Dim



Account Type - Dim

measures:

- Loan - Amount
- Interest

b) OLAP operation

[Customer, Branch, Time, Account - Types]

operations:

- * Roll - NO Customer → All
- * Roll - NO Account - Types - All
- * Roll - NO Time - year
- * Keep Branch

Result

[All, Branch, year, All]

c) Total cuboids

each dimension has 4 level

$$4 \times 4 \times 4 \times 4$$

$$= 256 \text{ cuboids}$$

4) Manufacturing company

1. Roll No

Summarize

* Defect Rate

* Product $\rightarrow$ category

* Time $\rightarrow$ current year

* Fact $\rightarrow$ All

Result:

[category, year, All, Factories]

2 Drill-Down

* Department

* Product type

2) Drill-down

Focus

* Electronic category

* specific Region

analyze:

category → subcategory

skew

measure:

- * Sales Revenue

- * Units sold.

slice

condition:

- * Gender = Male

- * membership = Bronze

Display:

- * Sales Revenue

Actions:

- * All stores

4) Dice

- * Specific cities

- * Selected product categories

- * Specific time periods

~~Analyze:~~

- * Sales Revenue

- * unit sold